AI ETHICS

Law and Regulation of AI: Global Perspective

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How do we govern the emergence of new technologies in Industry 4.0, specifically the rapid development of artificial intelligence in the past 20 years? In this module, I discuss the legal and regulatory aspects of AI from a global perspective. With the use of recent examples of new regulations on AI-based technologies, I will explain some of the approaches and challenges when it comes to regulating AI. I close with a brief overview of the state of global regulation at present and make the case for the adoption of international regulatory standards when it comes to governing AI.

Lecture Transcript

0:04 Hello, my name is Professor Dr Avinash Dadhich. I am from India. And I am working as a Professor of Law and Dean of the Law School in GLA University Mathura and I'm also running a center called Artificial Intelligence, Law and Society. And in my center, we try to examine how AI is affecting society and what law and regulation can play an important part to get the maximum positive aspects of artificial intelligence. At the same time, we also examine what type of challenges or threats which we are receiving or which may receive in future for this society. Today, I'm going to discuss artificial intelligence and its law and regulatory part from a global perspective.

1:03 So let's start. So, these are my email IDs. So, first one is my official one. And the second one is my personal one so you may contact me on these email ids.

1:44 So, as Victor Hugo said, “Nothing is more powerful than an idea whose time has come”. So without any doubt, we can say easily that the idea of artificial intelligence and other emerging technologies which are going to affect human life is already there. And as per this time for AI and life into 2030 report, so these are the areas where the human life is going to be greatly affected, you know, impacted, you can say, in terms of transportation service, robots, health education, law, resources, communities, public safety and security, employment and workplace and finally home service robots and entertainment. So you can see more or less that the most important aspects of human life will be impacted through AI in the next 10 years. So I think, you know, when I say that, this Stanford report, it says very clearly that yeah, it is there, it will be there, and it's going to enter in more and more domains of human life. So as you know, AI has seen a very interesting phase, winter to spring. And I know you're much aware of the little bit of the historical element of AI. So I won't go into that detail. But you can see that in the 1950s, and 60s, when scientists and innovators were talking about AI, it was more like an idea that how to develop and use AI to build high quality machines, okay, so there was less of an element of autonomy, but more like efficiency in existing machinery. However, no significant technology came out from this during this era. And finally, we can say in the 80s, that AI research and the entire thought process went to AI winter era. You know, people started saying that, okay, this is not going to very, you know, going to bring much change in our innovation spectrum. So it didn't work. But then again, I said that Victor Hugo, after the 1990s, and especially post 2000, the idea of AI started getting new momentum. And the reasoning behind this new momentum was that advancement in computing power, low cost of storing data, and high quality of digital data. So these are the three reasons that that really pushed the idea of AI into reality, and generated a new excitement in the public and private sector. So for example, if I just give you some data, you can see the cost of storing data has come down from a half million US dollar, a gigabyte in 1982 to two cents a gigabyte in 2017. Okay, so now maybe it's less than 2021. So you can see now you can't, you can't really believe what has happened in the last 20 years. In terms of technological growth.

5:05 So before moving to different theories of regulations relating to AI and emerging technologies, we need to understand a little bit about what AI is and what AI can do. So, AI can have different dimensions, but mainly we will talk about four dimensions. The first one is perception. This is the first level of AI, this is the beginning of AI, where the technology is used to collect and interpret the data, to understand the situation, and describe it in a more efficient manner. Okay, so we can use natural language processing, computer, visual and audio processing technologies. Okay, so for example, Apple's Siri, Microsoft's Cortana, Google's Assistant and Amazon's Alexa are good examples of this particular dimension. The second one is a prediction. This is where AI is helping companies to predict the behavior of their consumer to develop precisely targeted advertisements or services to them. So like, you can say Netflix is using algorithms that clearly suggests to us that what we should see and what are our likes and dislikes. So this is like a prediction, I'm just giving a very simple example. So even a layman can also understand that what AI is doing in our day to day life. Then the third one is that how AI is helping us to achieve a particular goal like a drug discovery, route planning, dynamic pricing, okay, so the good example can be Wealthfront - an AI supported online platform provides automated suggestions to consumers to maximize asset allocation and wealth management. So now AI is not only providing information, not only predicting but working in, in semi automatic manner, obviously, you give a goal and then AI is using its algorithms and other technologies to achieve a particular goal. Then the most sophisticated and that I think the most problematic one, also we can see integrated solutions - the most updated complex use of artificial intelligence is to its capability to work with other technologies. So first, you need to understand that AI is not itself a technology, but it's a bundle of technologies. Okay, you cannot say that, okay, we want to regulate AI, but at the same time, we ignore other technologies, other innovations, okay. So when I say robotics, blockchain, so these type of you know, technologies are also emerging. And these technologies work in an integrated manner. So like, for example, Baidu's or a Tesla, autonomous car functions in a controlled environment and it's integrated solution. So before moving to law and regulation, just talk about weak and general or strong AI. So the first one, the narrow, or the weak AI is based on simulated thinking which looks intelligent, but does not have an independent consciousness about its action and outcomes. Okay, so it doesn't have its own mind. It's, you know, just mimicry of human brain, or working in a very controlled manner. Okay. And normally limited to one or few given tasks. So like, for example, Google Maps, okay, or chatbots are good examples of this. Okay, they look like they're thinking, but they are not thinking they are working in a very controlled manner. Okay. So, the next one, which we are going to talk, I think there's the central idea next time, 15 years, the general AI - the real independent thinking, intelligence consciousness in the machines. Okay. So I think that will be a problem because when we'll talk about some lethal autonomous weapons, so then I think that those types of weapons if they are using general AI, they will take a call that to whom they should kill or not, okay. So when we give consciousness or when we give a decision making power to a machine or an AI, that I think the situation is a little bit different.

9:35 So first, we need to understand our goals and what we want to achieve, you know, because regulations are always very tricky business. Because obviously we will talk about the trade off of the regulations and different types of regulations, different theories, but first, we need to decide what we want to achieve through the law and regulation of AI. So, throughout history, throughout history, humans have shaped, developed and adapted to new path breaking technologies. So, this is not happening first time like okay the new way of technology, new generation of technology is coming and we have to adopt. We are doing it since I think 1000s of year. So, the measure of success of any existing and emerging technology is the value they create for quality and security of human life. So, we have seen many technologies in past where humans did not adopt, because those technologies were not bringing any quality or security for human life. So, these are the benchmarks basically to define and decide whether that particular technology should continue in human civilization or not. So, in order to achieve this goal, policymakers should design a legal and regulatory framework to enable people to understand these technologies properly. So, the first thing that because we are living in a democratic world, so, what we want transparency and accountability towards people. So, people must be able to understand the technology very well if there is too much secrecy or you know, less information available for the public that is not a democratic way. So, the first objective of legal and regulatory framework to create an environment where people can understand these technologies. Second, after understanding they must be able to participate in the utilization, okay. So, they must, they must be able to participate in the utilization, it's not be very selective, it should not be very selective, it should not be very classy, more and more people must be able to participate in that utilization process. And finally, the build trust between humans and machines. So, when I say more autonomous machines are coming now in 2021, maybe 25, 30 - we will see more and more. Autonomous machines are emerging in human civilization. So there is an issue. There is an issue of mistrust between machines and humans. Okay, so this mistrust can create a lot of confusions, initially for humans later on for the machines also, because machines, once they start thinking, independently, they become independent in their thinking process. So they will also observe human behavior. So what we don't that they should not consider humans as their enemy. So there has to be a trust between machines and humans the way we have developed trust between humans and animals. Okay, so I think this is in a new generation, where a new spice will come that animals, humans and machines, and they will have also their independent outlook, about life, about world, about the humans, about the environment, so we cannot ignore them, and they will not be our slave, but they will have their own independent consciousness. And that next one is the legal and regulatory approach should also help the society's adaption to these technologies. So that's very important, the law should be flexible and good enough to help the society to adopt those technologies in a proper manner, considering the new type of opportunities and challenges are presented before human civilization. So you need to understand law and regulatory bodies are basically gatekeepers of society, whenever something new happens, they come to know and they need to react, okay. They need to react that something is happening now what to do, how to protect the society, how to create maximum welfare in the society through those technologies.

13:54 So what's new in Industry 4.0? Because we have seen that Industry 3.0 We have adopted them very well. There were a lot of concerns of people that these technologies will destroy human civilization, but nothing happened. You know, still we are surviving. So what is new in this Industry 4.0? And the new is absence of human intelligence. So far from Industry 1.0 to 3.0, humans were always there, okay. Machines were working under the leadership and guidance and intelligence of humans. But now what we are seeing maybe in Industry 4.0 or say now, these are all jargons, Industry 5 0r 4.5, or Industry 5.0. But the idea is that now we are having more and more independent machines. And so we need to be a little bit careful, because less human intervention and more autonomy to machines brings a new challenge for human civilization. So regulatory mechanism - challenges - obviously when I don't want to talk about negative aspects of technology. It's a debatable yet there are a lot of issues. So I will talk all about the regulatory part right now that what could be the regulatory challenges. So the first regulatory challenge we can say that designing a robust legal and regulatory mechanism is a challenging task, as described by Judge Easterbrook in his famous paper titled “Cyberspace and the law of Horse”. He argues that billing an ex novo law for new technologies invites error in legislation. What he says that when the subject matter of law is emerging at a rapid and uncertain way, so his theory was that you cannot create ex novo law, it has to be ex post, that something has happened, and now the law should react. But I think, we will discuss later on, that this is not the situation in European Union and other countries also. The governments and regulatory bodies are adopting ex novo laws to regulate new technologies. The second challenge comes from the Harvard Law School approach, we say that the “overly rigid regulations to emerging path breaking technologies might stifle innovation process”. So obviously, we don't want to kill innovation. That is not the goal of any philosopher or any intellectual or any regulatory person. We want innovation, but at the same time, we want to regulate them. So the Harvard Law School says that when you overly regulate the technologies, innovation dies, okay, so these are the fundamental challenges when we start talking about regulation.

16:54 Then regulatory mechanism - challenges - the moral and ethical issues have swiftly called for an all out ban for a new AI production service, like lethal automated weapons and sex robots. So in when when I say about the law regulations, we have to include moral and ethical issues also. It cannot be only technological or economical, but moral and ethical issues of the society is equally important when we are trying to regulate the technology. But at the same time, when we include moral and ethical issues, despite having very good intentions, we also understand ethics and morals are very different in all countries. And unfortunately, now in 2021, or the last five, seven years, what we have seen that countries are having their own way of defining and interpreting morals and ethics at the international level. So this is a new challenge. So these complete restrictions are difficult to put on AI. Because see when I say, oh, let's you know, regulate the technology or put a complete ban on AI technology, why it's difficult. Four reasons. The first one is most of the research is out of the state’s domain. When you see in the last, especially after the Second World War, the states were very much active in developing technology, public laboratories were bringing a lot of new research, obviously private companies too but the public laboratories were very much active. But this is not the case for AI anymore. Geographically fragmented, but highly integrated through technology. So this AI is geographically fragmented - it is happening all over the world. But it's highly integrated through technology. When I say blockchain, you know, a cryptocurrency. So there is no geography in it, but it's very integrated through technology. Having multiple dimensions in terms of use, quite secretive in nature. So, when I say a particular technology, you can have one maybe ABC aspects of uses, but there can be D E F, also XYZ also. This is very difficult to understand which aspects we want to regulate, or which aspects we want to control. Okay. And the second one is very secretive in nature, what is happening in AI world right now, very few people are really aware of this thing in a real sense, you know. So most of the technologies are being developed in a very secretive manner. And finally, in the, in the absence of any global law, or institutions, hard to examine, and monitor the negative side of it. So unfortunately, we don't have any even national mechanism right now, forget about international mechanism. So it's very difficult to assess the positive or negative aspect of a technology at national as well as international level. We need something at international level because as I said geography doesn't matter anymore. The nation doesn't matter anymore. If a negative technology is being developed in country X, very quickly and easily that technology can move to a country like 10, 20, 30, 50 countries without any intervention of the government.

20:30 Now, let's talk about some regulatory trade offs. You know, when we start talking about regulations, what type of trade offs you are going to face. So very interesting example is that disabling regulations. Regulations can nip technology progress in the bud. Like, for example, a very interesting historical example is Red Flag Act of 1865, introduced by the UK Government. And as per this law, they introduced this one automobile industry. And this in this law says that maximum speed limit is two miles per hour in urban areas, and required a neutral person to carry a red flag 60 yards in front of the vehicle. You know, because people were very much afraid of automobile in the 1860s, 70s. The way we are afraid of emerging technologies right now. And the same thing, people were very much afraid. What is this? How it's happening? Maybe it's they will say the devil or something else. So and you know, so to avoid any type of accident 60 yards in front of the vehicle. Finally, what happened? It's very clearly observed that regulation stifled the development of British motor car industry for long 30 years. And then, so I think, you know, when you introduce too much regulations, finally, the industry suffers. Even close to digital innovations, the regulation of Bitcoin, payment, blockchain platforms, including caps on usage, and transparency requirements. So all these things are happening in terms of Bitcoin or blockchain technologies. They want to have, you know, names of the people, but the the very purpose of anonymity by the users, you know that the users wanted a anonymous system. That's why they created these technologies. And now the regulators want to disclose their names. So this can be a good example of disabling regulation. Knee-jerk regulations. So these type of regulations - inefficient over regulation in response to risk, incidents and accidents, this is very interesting. Sometimes, we just want to protect the society. Okay. And in that regard, we just want to create a law which makes sure that no accident, no risk, or nothing should happen negative in society. So this can be a problem. For example, we completely banned the cloning process. I am not in favor of cloning, but what I'm trying to say maybe something could come out from this technology, positive for the society. But just to stop this technology at the very early stage, UN adopted a Declaration on Human Cloning, which called upon member states, to adopt all measures necessary to prohibit all forms of human cloning. So this is one example that if you if you believe that this is too risky, you stop it at that level. So like, for example, if I take it another example is AI - airline autopilot. People are very much afraid at how it's possible. A human is not flying an airline, a machine is flying an airline. The reality is 40 to 50% time only machines are flying airlines, but very few. But people still believe that there are few pilots, you know, in the airplane. But what is the problem? The problem is that once we assume that AI can also operate an airline, and if something some accident happens, then few humans are still very much comfortable with accidents caused by human error, than a accident caused by a machine error. So in that situation, if something goes wrong with machines' decisions, and the demand would be of the society that let's stop it, we don't want to run it. Then the third one that's very interesting, that rent seeking and regulatory capture. So this is very important. You need to understand a lot of lobbying firms, interested parties, private interest group. They are always trying to capture the regulatory environment okay. So they want to create a regulation or they want to create a regulation for their own benefits instead of the social benefit. They want to do it for their own. Like, for example, the car dealer industry has sought enforcement of protective legislation against Tesla's B2C model. The taxi drivers and car companies have requested additional protection from the competitor competition of ride sharing apps like Uber. So, many times when you see that a particular lobby or a company or industry or a you know group where people are trying to do something or try. So you need to understand that they are not, they should not be able to capture the regulatory environment for their own benefit. Then regulatory timing is very important. So see that, as I said, in the first one (Judge Easterbrook), laws are very slow. So the same thing is regulation is a reactive process. When adopted, it may already late. So this is very interesting. Collingridge Paradox, it says that, "More generally, the regulatory process often struggles to keep abreast of technology. Emerging technologies present a 'quandry'; whilst a State would like to discourage research in harmful technologies". So the idea is that once we understand the technology, and we start taking a call, the technology is already changed. So we really don't know whether our regulations will have appropriate impact on that particular technology. Okay, especially this is happening, like, for example, in antitrust issues, in the technological regulations, even in pharmaceutical industry, this is very much happening. Then enabling regulation. So even free market spirited scholars claim that the absence of regulations can prevent technological evolution. Regulation is a necessity, because the alternative - a form of case by case litigation worse. So we want enabling regulations. Okay. So portal hypothesis, it says that the strict environmental, health and safety standards promote firms to improve their productivity and finds that properly designed standard can trigger innovation that may partially or more than fully offset the cost of complying with them. So when you have enabling regulations and companies integrate those regulations, okay, those laws, and then they start developing their innovation and their products, as per the regulations. So they are not against the regulation. But the Porter says that, once you introduce regulations on time, then companies start integrating them. So now let's move to the current state of AI regulation. So first to start with international organizations.

28:17 So I think this is the, the most document coming from United Nations. And as I said earlier, that we want documents, we want rules, regulations, treaties, conventions from the United Nation itself. Because the country cannot regulate AI. This is myth. It's very difficult to regulate AI and emerging technologies at regional and national level. It has to come from top level - the United Nations level. So "Recommendations on the Ethics of AI" was adopted by UNESCO General Conference at its 41st session on 24th November 2021. So a few days back, they adopted it. And this first global standard setting instrument is on the ethics of AI in the form of recommendation to provide AI with a strong ethical basis. The objective of this document is that it will not only protect but also promote human rights and human dignity. Okay, though it does not have any force of international law, but provides good grounds for future international law. So this is a nice document coming from United Nations, and though it doesn't have the force of law, but it's a good ground that where we can develop more convention and treaties in this area. The second one is very interesting that the United Nations Interregional Crime and Justice Research Institute established a specialized Center for AI and Robotics at Hague in 2017. And this center introduced a five year strategic plan covering from 2020 to 2025. And they will be covering these areas. So these are very comprehensive, like: building knowledge on the possible use of AI by criminals and terrorist groups; enhancing awareness of the threats programmatically generated; so like the voice content, such a deep fakes; fostering responsible AI innovation in the law enforcement communities. So basically, they are talking more from the law enforcement point of view, but because this is an UN, UN organ, and almost all countries are working very actively with this organization. So we can see that the ethical and responsible AI will be introduced in various member states through proper regulation. And this document, this strategic document can give good insight to all member states how to regulate their AI in cyber crimes or in terrorism, or in their day to day justice and other types of law enforcement mechanism. This is very important, like all are agreed, but still we are not having a law. So on December 2021, 2nd December 21, the group of governmental experts to the Convention on

Conventional Weapons will begin critical talks on whether to proceed with negotiation. Okay, so far 66 member states have called for a new legally binding framework for autonomy in weapons industry. But because a few members like Russia, Israel and the US, they are not very comfortable. They say that the creation is like too much premature, because these types of technologies are still evolving. And we don't want to kill them right now, as you see it, as I said that the United Nation killed human clone technology. And they are also trying to kill this technology also, because once this technology is on ground, and it's already on ground, in a very secretive manner, but once we have killer robots, it will be very difficult, very, very difficult to protect the human dignity, because then these robots and these technology weapon system will decide who should live, who should die without any human intervention.

32:45 So these countries, they believe that the existing international law, humanitarian law is sufficient to deal with these issues. However, most of the countries, they believe that this is not the case. These weapons are very different from our conventional weapons and we need to have a separate law to regulate these weapons and we should ban them. But as you know, politics of United Nations is still it's not there. But this is one area, which we really need to see in future that how it's going to evolve. Now, the EU, the I think the European Union, they took the leadership role in this area, and they introduced AI Act 2021. So this is the document which talks about in a very comprehensive manner in a very, very, very technical manner that how AI should be regulated in a country or in a region. So this regulation was introduced and released in April ‘21, and can be considered as a direct defy to US regulatory view that the law should leave emerging technologies alone. The proposal sets out a nuanced regulatory structure that bans some use of AI, heavily regulates high risk users and lightly regulates less risky AI system. So this is more technical document which talks about high risk as well as low risk AI systems. The proposal would require providers and users of high risk AI systems to comply with rules on data and data governance, documentation and report keeping, transparency and provision of information to users and human oversight, and robustness, accuracy and security. So these things. See, I think the important thing is a human oversight what we want it the human cannot take a plea that this was done by machines or by AI. Ultimately humans individually as well as their companies will be responsible for any breach of the law. This regulation also bans AI systems that cause or are likely to cause physical or psychological harm through the use of subliminal techniques or by exploiting different groups or persons due to the age, physical or mental disability. So, this is a complete ban. Still, we are not very clear because these are very open ended words, but it gives some right to member states how to regulate a particular technology, if that particular AI or a product or a service is creating some serious risk for some specific groups. And very interestingly, it prohibits AI systems from providing social scoring for general purpose by public institution, this is happening in China, so they don't want that any member states should follow the same path. It also precludes the use of real time remote biometrics identification systems such as facial recognition in publicly accessible space for law enforcement purposes. So this is very like a thin line balance where they are allowing member states to adopt facial recognition only for law enforcement purposes. And in 2021, the US Federal Trade Commission (FTC) also came up with a new guidelines aiming for truth, fairness and equity in your companies use for AI 202. And the main idea of the FTC that the company should comply and they should do these following these things like rely on inclusive data sets, company should think about the ways to improve the data set, design their models to account for data gaps, okay. And any manner it should not discriminate discriminate based on race, gender, or other protected class. Okay, so you can say the FTC has also issued a guidelines for the companies and what they want a transparent and independent manner. So that companies can choose how to do their business at the same time, people, the citizen, can also monitor them, and whether the truth, fairness and equity is happening or not.

37:36 So finally, I will stop here with my suggested regulatory framework. So the more appropriate regulatory approach is to examine this issue from the public policy, okay. When I say public policy is democracy, security, privacy, and a specific field of law, IPR, competition liability laws. So, instead of having very much technical analysis, we should see these issues from the public policy and domain specific issues. When I say AI generated art or AI generated intellectual property, okay, should we allow it or not? So, in that scenario, we should look into the fundamental concepts of IPR itself and try to find a solution. The same thing if happens in competition law, antitrust law, when companies are colluding with each other, they are making some anti competitive effects through technology, we should try to find solutions in competition law itself. However, when I say this legalistic approach or public policy approach, this approach cannot be developed only by legalistic approach, ignoring technological, social, socioeconomic and political conditions of a country or a community. See the public policy and the legalistic approach is a good option, but at the same time, we should also integrate technological, socio economic and political conditions of a particular country. One size does not fit to all when you are trying to regulate new emerging technologies. So when I say the emerging uses of robots can have a different social and economical considerations in Japan than in Africa, okay. So we cannot have one law, one regulatory framework to examine laws and regulations of emerging technologies. Okay. As a matter of fact, the present global political and economic order is moving back to the pre 1950 era, where individual member states had different views and solutions to global problems. This is very, very, very, very problematic when I say at the global level, we need some answers. But unfortunately now we have moved back to 1950 era where we all agree that problems are there, and we want to solve them at the global level, but they're, they have their own approaches to them, they have their own solutions. So I think this is very tricky situation because all challenges by artificial intelligence or other emerging technologies are not regional or national, but they are international. But unfortunately, we are not ready to find a common solution at the international level. The globalization of regulation of AI is very important as technologies and corporation evolved does not, don't expect or accept national boundaries. So this is very tricky, because all these big MNCs, big tech companies, they are working across the globe without worrying too much about the national laws. At the same time, the purpose and effect of AI also differs from developing world to developed economies. This is again very interesting. The same thing that climate change issues. Why we are having a lot of disputes, because the developing and developed countries are having different solutions, they have different ideas about this particular problem and solution. So same thing happens with the AI also. Therefore, these factors are equally crucial at a national and international level in designing laws and regulations for AI. So in summary, I can say that we all want to regulate artificial intelligence and emerging technologies. But when once we start making rules and regulations, then we need to integrate social, economic, political, technological dimensions of a particular country. But at the same time, this is the time when we must have some international laws on these technologies. So with these words, I say thank you. Thank you very much giving me this opportunity. And if you have any question or any doubt, please write me an email. I will be very happy to answer you. Thank you very much.